شـركة غـاز البصـرة Basrah Gas Company	**Basrah Gas Company** **Safwan Road** **Khor Al Zubair** **Basrah, Republic of Iraq**	**WO-174** **Framework Agreement** **Dimensional Control Survey**

SAP Outline Agreement No. **4610069586**	Contract No.: **CW38917**		
SAP PO No.:	Contract:	Effective Date: **01 Jan 2024**	Expiry Date: **31 Dec 2026**
CONTRACTOR: **Fugro Middle East**	WO Effective Date: **Upon Receipt of PO**		
WO Title: **WO-174**	Scheduled Completion Date **Mobilisation – Week upon receipt of PO** **Fieldwork – 1 week from mobilization completion** **Reporting – 1 week from fieldwork completion**		
WBS Account: **O.PD.KAZ.COM.001.AC000**	WO VALUE: **$ 28,620.00**		
WO Reference: **WO-174**	Pricing Basis: **Re-measurable**		

This WORK ORDER (WO) is subject to the terms and conditions of the CONTRACT referenced above. In case of conflict between the terms set out hereunder and the terms of the CONTRACT, the terms the CONTRACT shall prevail.

This WORK ORDER comprises the following attachments:

1. Scope of Works and Deliverables
2. Timeline and Key Dates
3. Schedule of Prices

Technical Proposal – 3D Scanning and Dimensional Control Survey for Sphere 61-S-01A Isolation (IZOMAX AOGV)

Client	Basrah Gas Company (BGC)
Asset / Location	KAZ (Khor Al Zubair) NGL – Sphere 61-S-01A and associated ESDVs
Work Order / WBS	WO-174 / O.PD.KAZ.COM.001.AC000
End User of Data	IZOMAX (AOGV isolation tool design) and BGC Mechanical Static
Prepared by	Nelson Munyamba – Services Technical Manager, Fugro
Revision / Date	Rev 00 – 07 October 2026

1. Introduction

BGC intends to isolate Sphere 61-S-01A at the KAZ NGL Plant using IZOMAX Add-On Gate Valve (AOGV) isolation equipment, installed at the flanged connections of the sphere inlet and outlet ESDVs. To design and fabricate this specialised mechanical equipment, IZOMAX requires highly detailed, dimensionally accurate as-built geometry of each flange set and its immediate surroundings.

Existing terrestrial laser scan data of the area (acquired in November 2024 with a FARO scanner) has been reviewed by BGC and Fugro and does not achieve the level of detail specified by IZOMAX. Dedicated scanning of the identified flanges is therefore required using a metrology-grade handheld scanner.

Fugro proposes a metrology-led approach in which the **Creaform Go!SCAN handheld 3D scanner is the primary data acquisition system** for every flange set. A **FARO terrestrial laser scanner and a Leica total station are deployed as supplementary systems** to document the site, capture the surrounding constraints, and provide independent, redundant measurements used to validate the critical dimensions and clearances extracted from the Go!SCAN data.

2. Objectives

- Acquire sub-millimetre, high-density 3D scan data of the seven (7) scanning points defined by BGC at Sphere 61-S-01A (Section 3.1) using the Creaform Go!SCAN.
- Extract the flange, bolting and interface dimensions required by IZOMAX for AOGV tool selection, design and fabrication.
- Document the surrounding pipework, supports, platforms and obstructions to define the available installation envelope and access constraints for each AOGV.
- Independently validate key Go!SCAN dimensions against FARO and total station measurements, so that every deliverable carries a documented, traceable accuracy.
- Deliver validated geometry packages, including Creaform native files (*.CSF), meshes and supporting reports, in a format directly usable by IZOMAX.

3. Scope of Work

3.1 In-Scope Scanning Points

The scanning points below are as defined in the BGC scope of work (R. Hatt, 04 October 2026) and the associated mark-up P&ID and 3D model extracts.

No.	Tag / Item	Size	Rating	Location / Service
1	ESDV-014	6″	Class 300#	Sphere outlet
2	ESDV-015	24″	Class 300#	Sphere outlet – 24″-P61507 header
3	ESDV-011	20″	Class 300#	Sphere inlet – 20″-P61508 header
4	ESDV-012	18″	Class 300#	Sphere inlet
5	MV807	2″	Class 300#	Inlet bypass
6	Flange set A	24″	Class 300#	Additional flange set – location to be confirmed by BGC/IZOMAX at site
7	Flange set B	24″	Class 300#	Additional flange set – location to be confirmed by BGC/IZOMAX at site

Each scanning point includes both flanges of the connection, the bolting, the valve body/flange interface and a local zone of approximately 1.5 m around the joint, so the installation envelope and clashes can be checked. Additional points (e.g. other ESDVs shown on the BGC 3D model mark-ups) can be added on a re-measurable basis at the rates in Attachment 1.

3.2 Activities

- **Pre-mobilisation:** method statement, risk assessment/JSA, HSE plan, equipment list and calibration certificates submitted for BGC approval.
- **Site visit (recommended):** a short pre-survey visit with BGC and IZOMAX to confirm the exact flange locations, access and scaffolding needs, insulation removal and permit requirements.
- **Mobilisation and permits:** site induction, permit-to-work and gas testing coordinated with BGC.
- **Control:** establishment/verification of local survey control and targets with the total station, tied to the plant coordinate system where available.
- **Primary acquisition:** Creaform Go!SCAN high-density scanning of all seven scanning points.
- **Supplementary acquisition:** FARO terrestrial scans and HDR imagery of the scanning areas, plus total station check measurements on key features.
- **Processing:** Go!SCAN meshing and dimension extraction; FARO registration; integration of all data in a common coordinate frame.
- **Validation and QA/QC:** cross-comparison of Go!SCAN dimensions against FARO and total station data; discrepancy reporting.

- **Reporting:** Daily Progress Reports (DPRs), Flange Detail Reports and a Final Survey Report.

4. Equipment and Methodology

4.1 Equipment Roles

Equipment	Role	Purpose
Creaform Go!SCAN handheld 3D scanner (VXelements software)	Primary	Sub-millimetre capture of flanges, studs/nuts, gasket gaps, valve-to-flange interfaces and local geometry for AOGV tool design.
FARO Focus terrestrial laser scanner	Supplementary	Site documentation: wider-area point cloud and HDR imagery of pipework, supports, platforms and access routes; installation envelope and clash context; redundant data for dimension validation.
Leica total station and survey targets	Supplementary	Survey control, scale and orientation; independent check measurements of centrelines, elevations and flange-face positions; tie to plant coordinates.
Hand tools (calibrated vernier calipers, steel tape)	Check	Spot checks of directly accessible items (e.g. stud diameter, flange thickness) to confirm scan-derived values.

4.2 Primary Acquisition – Creaform Go!SCAN

- Each flange set is scanned with the Go!SCAN, capturing flange outside diameters and thicknesses, raised faces, bolt holes, studs and nuts, the gasket gap and the valve-to-flange interfaces at sub-millimetre resolution.
- Positioning targets are applied around each joint to control drift on the larger 18″–24″ connections. Where needed, the target network is coordinated by total station to fix scale and orientation.
- The handheld system reaches occluded and congested areas (behind actuators, under pipe and close to supports) that a tripod scanner cannot see.
- Data is reviewed for completeness in real time in VXelements before leaving each point, which reduces the risk of returning to rescan.
- Outputs per point: native Creaform files (*.CSF), a watertight or surface mesh (STL/OBJ) and extracted dimensions.

4.3 Supplementary Acquisition – FARO Terrestrial Laser Scanner

- Multi-station scans around the sphere inlet and outlet areas document the as-built context: adjacent pipework, supports, platforms, handrails, cable trays and access routes.
- This data defines the space available to bring in, position and operate the AOGV equipment. It also identifies obstructions that would not be visible in the local Go!SCAN data.

- HDR panoramic imagery provides a photographic record of each location for IZOMAX and BGC reviewers.
- The registered FARO cloud is an independent dataset against which Go!SCAN centrelines, flange positions and separations are compared.

4.4 Supplementary Control and Checks – Leica Total Station

- Local control is established around the work areas and tied to existing plant control/benchmarks where available. Station descriptions and coordinate lists are provided.
- Total station observations to Go!SCAN positioning targets and FARO targets bring all datasets into one common, scaled coordinate frame.
- Independent check measurements are taken on key features (flange faces, pipe centrelines, elevations) to validate the dimensions extracted from the scan data.

4.5 Data Integration, Validation and QA/QC

Go!SCAN meshes are registered into the total station-controlled frame together with the FARO point cloud. Critical dimensions are extracted from the Go!SCAN data and checked against the redundant FARO, total station and hand-tool measurements, using the targets below:

Parameter	Primary source	Validation target
Flange/bolting detail (OD, thickness, PCD, stud size)	Go!SCAN	Agreement with hand-tool checks within ±1 mm
Flange-face separation / gasket gap, misalignment	Go!SCAN	Agreement with independent check within ±1–2 mm
Centreline positions and elevations	Go!SCAN + total station	Agreement with total station within ±3 mm
Installation envelope / clearances	FARO (context) + Go!SCAN	Agreement between datasets within ±5 mm

Any difference above these targets is investigated and, where necessary, the point is rescanned before demobilisation. Registration residuals, comparison results and deviation colour maps are reported in the QA/QC report. Final achievable values are confirmed in the method statement once site conditions (access, insulation, vibration and temperature) have been verified.

4.6 Safety and Operations

- All work is done under BGC permit-to-work, with continuous gas monitoring where the area classification requires it. The scanners and total station are not intrinsically safe and will be used only under an appropriate (hot work/gas-tested) permit.
- Scanning is non-intrusive and carried out with the plant live. No process shutdown is required, and safe distances from operating equipment are maintained.

- Work at height uses scaffolding/platforms provided by BGC and inspected before use.
- Fugro personnel will attend all BGC inductions, toolbox talks and HSE requirements, and will coordinate daily with the BGC area authority and IZOMAX.

5. Technical Benefits of the Approach

- **Fit for purpose:** the Go!SCAN delivers the sub-millimetre flange detail that IZOMAX requires and that tripod-based scanning alone cannot provide.
- **Validated, traceable dimensions:** redundant FARO and total station measurements independently confirm the critical dimensions, reducing the risk of a mis-fitting AOGV tool.
- **Complete site picture:** FARO documentation captures the access routes, clearances and obstructions needed to plan safe installation.
- **Reduced field time and rescan risk:** real-time completeness checks and a common control frame let all data be acquired in one mobilisation.
- **Proven on BGC assets:** Fugro has delivered multiple laser scanning projects for BGC facilities (see Attachment 2, BGC WO-168).

6. Deliverables

No.	Deliverable	Format	Content
1	Go!SCAN scan data – 7 scanning points	*.CSF (native)	Raw/processed Creaform datasets for each point, as requested by IZOMAX.
2	Meshes of flanges, valves and interfaces	STL / OBJ	High-resolution mesh per scanning point for tool design.
3	Flange Detail Report (per point)	PDF + XLSX	Flange OD/thickness, raised face, PCD, bolt-hole count/size, stud and nut sizes and lengths, gasket gap, face separation, misalignment/rotation, annotated views and photos.
4	Site documentation point cloud	E57 / RCP	Registered FARO point cloud of the scanning areas with HDR imagery.
5	Integrated review model	Navisworks NWD	Go!SCAN meshes and FARO cloud combined for review and clash checks of the AOGV envelope.
6	Control and validation report	PDF	Control coordinates, station descriptions, registration residuals, Go!SCAN vs. FARO/total station comparisons and deviation maps.
7	DPRs and Final Survey Report	PDF	Methodology, daily log, NPT/delays, personnel and equipment, calibration certificates.

7. Submission Documents

In line with the BGC/IZOMAX requirements, Fugro will provide the following before mobilisation:

- Method statement and HSE plan, including risk assessment/JSA.
- Site activity needs list and a responsibility matrix (Fugro / BGC / IZOMAX).
- List of scanning equipment and accessories with calibration certificates.
- Scanning programme and manpower plan.
- Technical specifications and data sheets for the Creaform Go!SCAN, FARO scanner and Leica total station.
- Personnel competency records and an induction schedule.

8. Programme and Timeline

- **Mobilisation (within 1 week of PO):** method statement approval, mobilisation, induction and permits.
- **Fieldwork (approx. 5 working days):** Day 1 permit-to-work, control and targeting. Days 1–5 Go!SCAN scanning of the 7 points. Days 2–4 FARO site documentation and total station checks (in parallel).
- **Processing and QA/QC:** Go!SCAN meshing and dimension extraction, FARO registration and cross-validation, running in parallel with the later fieldwork.
- **Reporting (within 1 week of fieldwork completion):** Flange Detail Reports, final report and data transfer to BGC/IZOMAX.



Figure 1 – Indicative programme (working days).

9. Resources

- 1 × Metrology / 3D Scanning Specialist – Creaform Go!SCAN acquisition and processing.
- 1 × Senior Surveyor – total station control, FARO scanning and validation checks.
- Office support – data processing, QA/QC and reporting, supervised by the Fugro Services Technical Manager.

10. BGC / IZOMAX Support Requirements

- Permits-to-work, gas testing, area authority sign-off and escorts as required.
- Safe access to all scanning points, including scaffolding where needed. Removal of insulation, guards or coverings from flanges to be scanned (scanning captures visible surfaces only).
- IZOMAX to confirm, before mobilisation, the critical dimensions and tolerances required for AOGV design. IZOMAX to attend site, or confirm at the site visit, to agree each scanning point.
- BGC to confirm the exact location of the two additional 24″ flange sets.
- Existing plant control coordinates/benchmarks and relevant drawings (P&IDs, isometrics, 3D model extracts) to be made available.

11. Assumptions and Exclusions

- Continuous access to the scanning areas during agreed working hours; minimal obstructions during scanning windows.
- Fugro will use existing plant control coordinates/benchmarks where available, or will be permitted to establish new ones.
- No process shutdown is required; scanning is carried out within safe operating envelopes.
- Scanning captures visible, accessible surfaces only. Concealed surfaces (e.g. gasket faces, internal bores, areas under insulation left in place) are excluded.
- The pricing is re-measurable: additional scanning points, extra field days or standby caused by permit or access delays outside Fugro's control will be charged at the rates in Attachment 1.
- Conversion of meshes to parametric CAD (STEP/STP) or 3D modelling of the AOGV tool is excluded unless agreed separately.
- Vehicle provision for the BGC representative (2 days per week) is not included.

12. Commercial

- Work Order value: USD 28,620.00 (re-measurable), as detailed in Attachment 1 – Pricing Schedule.
- Terms and conditions in accordance with Contract CW38917 / SAP Outline Agreement 4610069586.
- Validity: 30 days from the proposal date.

13. Closing

Fugro appreciates the opportunity to support BGC and IZOMAX with the isolation of Sphere 61-S-01A. Using the Creaform Go!SCAN as the primary system, with independent FARO and total station validation, gives IZOMAX the detailed, verified geometry needed to design and install the AOGV equipment safely and first time right. We look forward to arranging the site visit and to discussing this proposal with BGC.

Attachment 1: Pricing Schedule
Attachment 2: Sample Project – BGC WO-168

WORK ORDER Signature page

In witness whereof COMPANY and CONTRACTOR hereby execute this WORK ORDER and sign it as follows;

Signature: For COMPANY:	Signature: For CONTRACTOR:
Name:	Name: **Ahmad Yahya**
L1 Director	Title: CONTRACTOR REPRESENTATIVE
Date:	Date: **18-December-2025**

Signature: For COMPANY:
Name:
COMPANY REPRESENTATIVE
Date:

BGC PM	
BGC CE	